Case Study 7: Food Waste Monitoring System

1. Problem Statement

Develop a Python program to monitor food waste over multiple days. The program should accept the number of meals prepared and meals served each day, calculate the leftover meals and wastage percentage, and identify days when food wastage exceeds the acceptable limit.

2. Objectives

- To accept data from the user.
- To understand loops and conditional statements.
- To calculate leftover meals and wastage percentages.
- To identify days with excessive food waste.

3. Assumptions

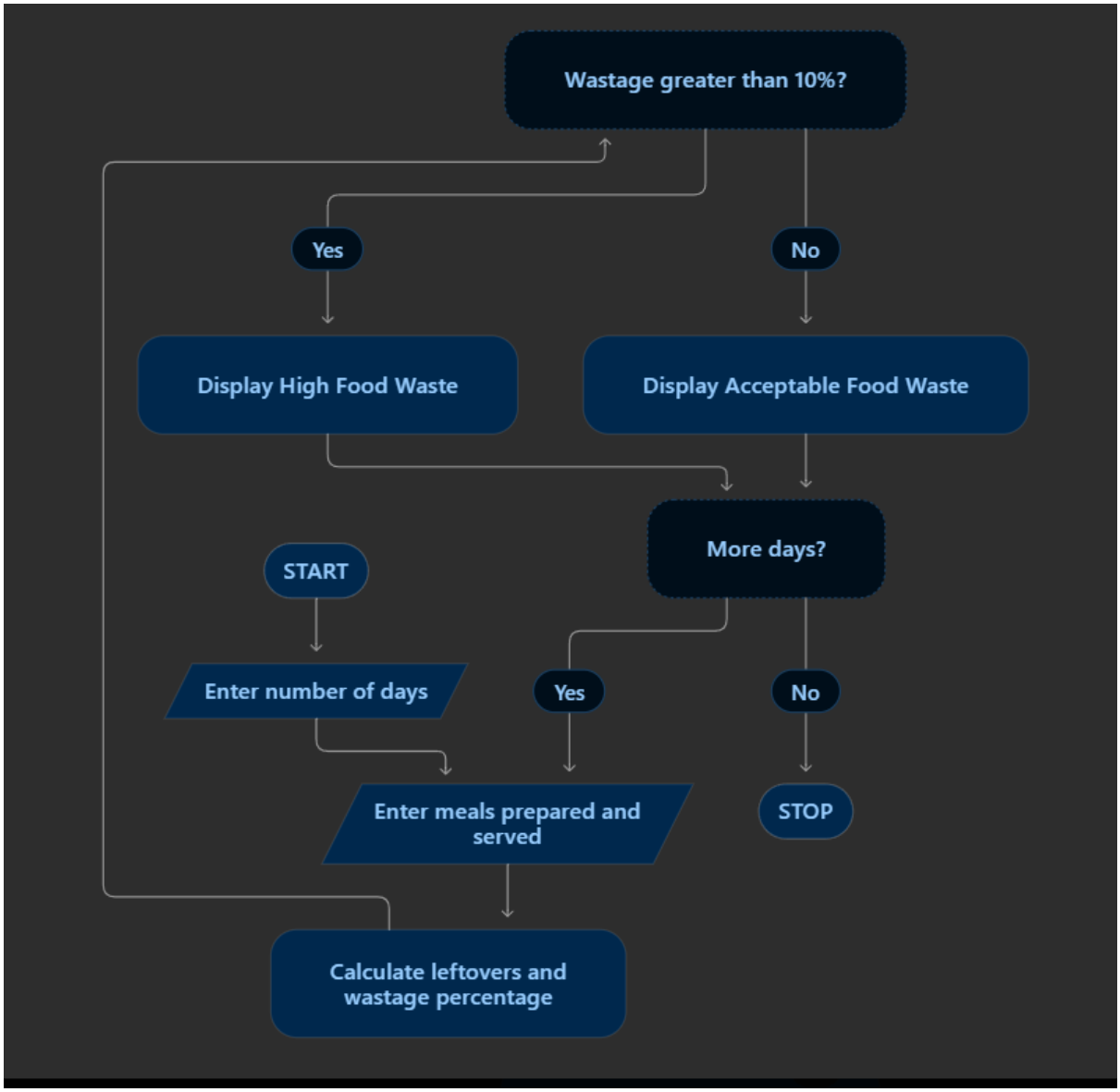
- The user enters the number of days to monitor.
- Meals prepared must be greater than zero.
- Meals served cannot be negative or greater than meals prepared.
- Wastage percentage is calculated using meals prepared.
- The acceptable wastage limit is assumed to be 10%.

4. Algorithm

1. Start.
2. Input the number of days.
3. Check whether the number of days is positive.
4. For each day, input meals prepared and meals served.
5. Validate the entered values.
6. Calculate leftover meals by subtracting meals served from meals prepared.
7. Calculate the wastage percentage.

8. If wastage exceeds 10%, display a warning.
9. Otherwise, display that wastage is within the acceptable limit.
10. Stop.

5. Flowchart



6.Pseudocode

START

INPUT number_of_days

```
IF number_of_days <= 0 THEN
   PRINT "Invalid number of days"

ELSE
   FOR each day

      INPUT meals_prepared
      INPUT meals_served

      IF meals_prepared <= 0 OR
        meals_served < 0 OR
        meals_served > meals_prepared THEN

         PRINT "Invalid meal data"

      ELSE
         leftovers = meals_prepared - meals_served

         wastage_percentage =
            (leftovers / meals_prepared) * 100

         PRINT leftovers
         PRINT wastage_percentage
```

```
        IF wastage_percentage > 10 THEN
          PRINT "High Food Waste"
        ELSE
          PRINT "Acceptable Food Waste"
        END IF
      END IF

    END FOR
END IF

STOP
```

7.Source code

food waste monitoring system.py

```python
days = int(input("Enter number of days: "))

if days <= 0:
    print("Invalid number of days!")

else:
    for i in range(days):
        print("\nDay", i + 1)

        prepared = int(input("Enter meals prepared: "))
        served = int(input("Enter meals served: "))

        if prepared <= 0 or served < 0 or served > prepared:
            print("Invalid meal data!")

        else:
            leftovers = prepared - served
            wastage = (leftovers / prepared) * 100

            print("Leftover meals:", leftovers)
            print("Wastage percentage:", round(wastage, 2), "%")

            if wastage > 10:
                print("High Food Waste")
            else:
                print("Acceptable Food Waste")

```

8.Test cases

prepared	served	Expected output
100	95	5%- Acceptable
100	80	20%- High food waste
100	120	Invalid meal data

9.Output

```
"C:\Users\tejas\PycharmProjects\Python Activity\.venv\Scrip
Enter number of days: 1

Day 1
Enter meals prepared: 100
Enter meals served: 95
Leftover meals: 5
Wastage percentage: 5.0 %
Acceptable Food Waste

Process finished with exit code 0
```

```
"C:\Users\tejas\PycharmProjects\Python Activi
Enter number of days: 2

Day 1
Enter meals prepared: 100
Enter meals served: 85
Leftover meals: 15
Wastage percentage: 15.0 %
High Food Waste

Day 2
Enter meals prepared: 100
Enter meals served: 40
Leftover meals: 60
Wastage percentage: 60.0 %
High Food Waste

Process finished with exit code 0
```

10. Conclusion

The Food Waste Monitoring System calculates leftover meals and food wastage percentages for multiple days. It uses user input, loops, arithmetic operations, and conditional statements to identify days with excessive food waste.